

Discovery of Finite-Order Resonance in Iterated Exchange Scattering

Identifying Sharp Peaks near Fine-Structure-Constant Inverse Values 137 and 128 with a Reproducible Wave-Packet Model

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Abstract

This study began with the explicit aim of testing whether an exchange-scattering model of a closed phase system could explain values associated with the low- and high-energy fine-structure constant. In the preceding study, the exchange-scattering coefficient R was swept in System A, which reads localization transfer between wave packets, and System B, which reads a white-cat/black-cat/gray-cat metastable interface. Sharp peaks were observed near the regions corresponding to $\alpha^{-1} \approx 137$ and $\alpha^{-1} \approx 128$, but the reason why peaks formed at those positions remained unresolved.

This paper reports a reanalysis of the full-range sweep, root-centered precision sweeps, operator-eigenvalue analysis, and 50- and 80-digit calculations performed to identify the cause. The analysis shows that the observed peaks do not arise from the assumed electromagnetic-coupling readout alone. They are generated by exact recurrence of the two-channel exchange-scattering operator

$$U_R = \begin{pmatrix} r & t \\ t & r \end{pmatrix},$$

whose antisymmetric eigenvalue returns to unity after a finite number of iterations:

$$U_R^n = I.$$

The finite-order roots are

$$R_{n,m} = \cos^2\left(\frac{\pi m}{n}\right).$$

The principal low-energy peak is $R_{124,23}$, and the former high-energy principal candidate is $R_{122,23}$. A separate even-order root $R_{620,117}$ lies near the high-energy value $\alpha^{-1}(M_Z^2) = 128.946 \pm 0.015$ and gives

$$N(R_{620,117}) = \frac{4\pi}{(1 - R_{620,117})^2} = 128.947864735670559.$$

Its difference from the quoted value is 0.0018647, or 0.1243σ .

The 50-digit calculations produced depths near 52, while the 80-digit calculations produced depths from 82 to 83. At an exact finite-order root, the residual vanishes in ideal arithmetic, so the depth is formally infinite. This is not a divergence of energy or amplitude; it means that the resonance line becomes infinitely sharp.

The principal result is not only the analytic root formula. Two numerical models originally constructed for different purposes—wave-packet localization transfer and observation selection at a metastable two-state interface—share the same complex exchange-scattering kernel and reproduce operator-intrinsic finite-order resonance. Exact recurrence can therefore be tested as an iterated wave-packet interaction calculation rather than remaining only an abstract operator statement.

After this analysis, a literature search revealed connections to exact state-independent Floquet recurrence from cyclotomic spectra, Farey organization of resonance diagrams, the decomposition of E_8 into 31 orthogonal eight-dimensional subspaces, and studies relating exceptional algebras to the fine-structure constant. These

works did not motivate the numerical experiment; they were identified only after the finite-order roots had been found.

We initially expected this numerical experiment to explain α . What we directly found was a different finite-order resonance phenomenon connected to a generally known class of exact recurrences. It remains unresolved why such sharp finite-order roots occur near the two physical values of α^{-1} . Determining whether this proximity is an arithmetic coincidence or follows from four-dimensional discrete geometry or another structural necessity is the central task of the next stage.

Keywords: fine-structure constant, exchange scattering, finite-order resonance, quantum recurrence, Floquet operator, wave-packet localization, metastable state, E_8 , reproducible computation

1. Research Motivation

1.1 A Search Deliberately Aimed at the Fine-Structure Constant

The motivation of this study must not be obscured. We explicitly expected that the exchange-scattering structure of a closed phase system might explain a value corresponding to the fine-structure constant.

We define the exchange weight by

$$G_R := 1 - R$$

and assume the correspondence

$$G_R = e$$

with the electromagnetic coupling amplitude in the standard theory,

$$e = \sqrt{4\pi\alpha}.$$

This is the correspondence hypothesis of the present study. We further define the readout

$$N(R) := \alpha^{-1}.$$

Since

$$(1 - R)^2 = 4\pi\alpha,$$

we obtain

$$\boxed{N(R) = \frac{4\pi}{(1 - R)^2}}.$$

Under this correspondence hypothesis, exchange-scattering coefficients can be assigned to the low-energy value

$$\alpha^{-1}(0) = 137.035999177(21)$$

and the high-energy value

$$\alpha^{-1}(M_Z^2) = 128.946 \pm 0.015.$$

Thus, the region $R \approx 0.7$ was not selected accidentally. We examined whether the effective-response region found in the localization-transfer and metastable-interface experiments could connect to the exchange weight corresponding to the electromagnetic coupling amplitude.

1.2 System A: Wave-Packet Localization Transfer

The first starting point is a localization-exchange model in which an exchange-interference scattering matrix acts on wave-packet vectors [1,2]. A is a low-localization wave and B is a localized wave containing additional harmonics. We iterate

$$\begin{pmatrix} A' \\ B' \end{pmatrix} = U_R \begin{pmatrix} A \\ B \end{pmatrix}.$$

The readouts are

$$L, \quad N_{\text{eff}}, \quad B_{\text{to}A},$$

where L is localization, N_{eff} is the effective harmonic order, and $B_{\text{to}A}$ measures the transfer of the initial B-side harmonic distribution into A.

At the endpoints of complete transmission $R = 0$ and complete reflection $R = 1$, the distinction between the low- and high-order waves was preserved. Under intermediate scattering, localization and harmonic structure were periodically transferred between the two channels. In particular, the localization and effective harmonic orders of the two channels approached one another near $R \approx 0.70$.

This model was not originally designed to search for exact recurrence. It was constructed to numerically investigate an interaction in which a wave packet appears to become localized through exchange with another wave packet.

1.3 System B: White-Cat, Black-Cat, and Gray-Cat Metastable Interface

The second starting point is the white-cat/black-cat/gray-cat model, which reads the distribution and observation selection of two states A and B [3]. From complex amplitudes a and b , define

$$p_A = |a|^2, \quad p_B = |b|^2, \quad S = p_A - p_B.$$

- $S \approx +1$: white-cat state A.
- $S \approx -1$: black-cat state B.
- $S \approx 0$: gray state.

Among gray states, a state that fluctuates with small amplitude, survives a weak C readout, and is selected into A or B by a strong D observation is called a gray metastable phase.

This model was likewise not designed from the outset to reproduce the fine-structure constant or Floquet recurrence. It was constructed to distinguish a readout that does not destroy a metastable mixed state from strong observational selection.

1.4 The Question Left by the Previous Paper

The paper of July 15, 2026 [4] connected Systems A and B through their common exchange-scattering coefficient R . It found that:

- with an additional harmonic, the response concentrates in the region corresponding to 137;
- the candidate position barely moves when the harmonic type, amplitude, phase, or wavelength is changed; and
- the full System B sweep contains deep peaks near the regions read as 137 and 128.

The following question, however, remained unresolved:

Why do sharp peaks form at those positions?

This paper reports the investigation undertaken to answer that question and the resulting identification of the cause.

2. Computational Model and Readout

2.1 Exchange-Scattering Coefficients

For a reflection probability $R \in [0, 1]$, define

$$\theta(R) := \arcsin \sqrt{R}.$$

The transmission and reflection amplitudes are

$$t = e^{i\theta} \cos \theta, \quad r = -ie^{i\theta} \sin \theta.$$

Therefore,

$$|t|^2 = 1 - R, \quad |r|^2 = R.$$

2.2 Two-Channel Exchange Operator

One scattering iteration is defined by

$$\begin{pmatrix} A_{j+1} \\ B_{j+1} \end{pmatrix} = U_R \begin{pmatrix} A_j \\ B_j \end{pmatrix}, \quad U_R = \begin{pmatrix} r & t \\ t & r \end{pmatrix}.$$

Because the coefficients satisfy

$$|r|^2 + |t|^2 = 1, \quad r^*t + t^*r = 0,$$

we have

$$U_R^\dagger U_R = I.$$

Thus, the exact fixed- R map is unitary and does not dissipatively attract the state norm to a point.

The “concentration’’ at a peak in this study is not dissipative contraction in state space. It is resonance selection in R -parameter space, where a closed-orbit condition concentrates at particular values. The increase of wave-packet localization observed in System A is a redistribution of localization by unitary interference, not contraction of the total norm.

2.3 Initial State of System B

Let

$$A_0 = \sqrt{\frac{1}{2} + s_0}, \quad B_0 = \sqrt{\frac{1}{2} - s_0} e^{i\phi}.$$

The principal sweep uses

$$s_0 = 0.01, \quad \phi = 0.$$

The initial population difference is therefore

$$S_0 = |A_0|^2 - |B_0|^2 = 0.02.$$

2.4 Gray Error and Depth

At each time, define the population difference by

$$S_j = \frac{|A_j|^2 - |B_j|^2}{|A_j|^2 + |B_j|^2}.$$

For a prefix of length k , define

$$\bar{S}_k = \frac{1}{k} \sum_{j=0}^{k-1} S_j,$$

$$A_k = \frac{\max_{j < k} S_j - \min_{j < k} S_j}{2},$$

and

$$D_k = |\bar{S}_{k,\text{second}} - \bar{S}_{k,\text{first}}|.$$

The gray error is

$$\varepsilon_k = |\bar{S}_k| + |A_k - 0.02| + D_k,$$

and the depth is

$$d_k = -\log_{10} \varepsilon_k.$$

A periodic orbit becomes a deeper peak when its mean approaches zero, its amplitude approaches 0.02, and the drift between the first and second halves approaches zero.

3. Full-Range Sweep in the Previous Study

3.1 Sweep Conditions

Using the minimal System B implementation v5, we uniformly swept 158,625 points over

$$R_{\min} = 0.68660290255614798,$$

$$R_{\max} = 0.70246536756843059,$$

with

$$\Delta R = 10^{-7}.$$

The same scattering update, gray error, and stopping condition were applied to every R . The sweep evaluation formula contains neither α , 137, 128, nor a finite-order root.

3.2 Full-Range Sweep Figure

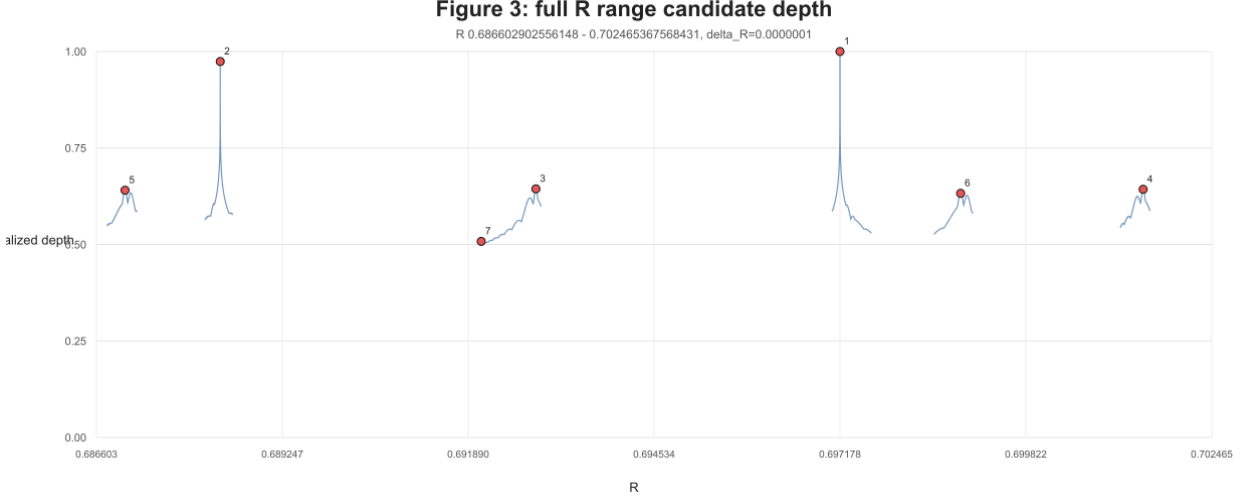


Figure 1. Full System B sweep obtained on July 15, 2026. The horizontal axis is R , and the vertical axis is depth normalized within the candidate bands. The deepest candidate 1 appeared at $R = 0.697177902556148$, and candidate 2 appeared at $R = 0.688363902556148$.

The full-range figure showed that the response was not a single peak but a collection of narrow candidate bands. At that time, candidate 1 was read as a peak corresponding to the low-energy value $\alpha^{-1}(0) \approx 137$, while candidate 2 was read as a peak near the high-energy value $\alpha^{-1}(M_Z^2) \approx 128$.

However, a spacing of 10^{-7} could not determine the true center, width, limiting depth, or generating mechanism of each peak.

3.3 Original Candidate Values

Candidate	R in the full sweep	$N(R)$	Interpretation at the time
1	0.697177902556148	137.036020287643	Near low-energy 137
2	0.688363902556148	129.394062925467	Candidate near high-energy 128

Candidate 2 was not itself 128.946. Nevertheless, because it was the second principal peak in the same full-range figure, its origin required identification.

4. Root-Centered Precision Sweeps

4.1 Low-Energy Side: $R_{124,23}$

Candidate 1 from the full sweep was magnified while reducing the spacing successively to

$$10^{-8}, \quad 10^{-10}, \quad 10^{-12}.$$

The peak center converged to

$$R_{124,23} = \cos^2 \left(\frac{23\pi}{124} \right) = 0.697177927556659290 \dots$$

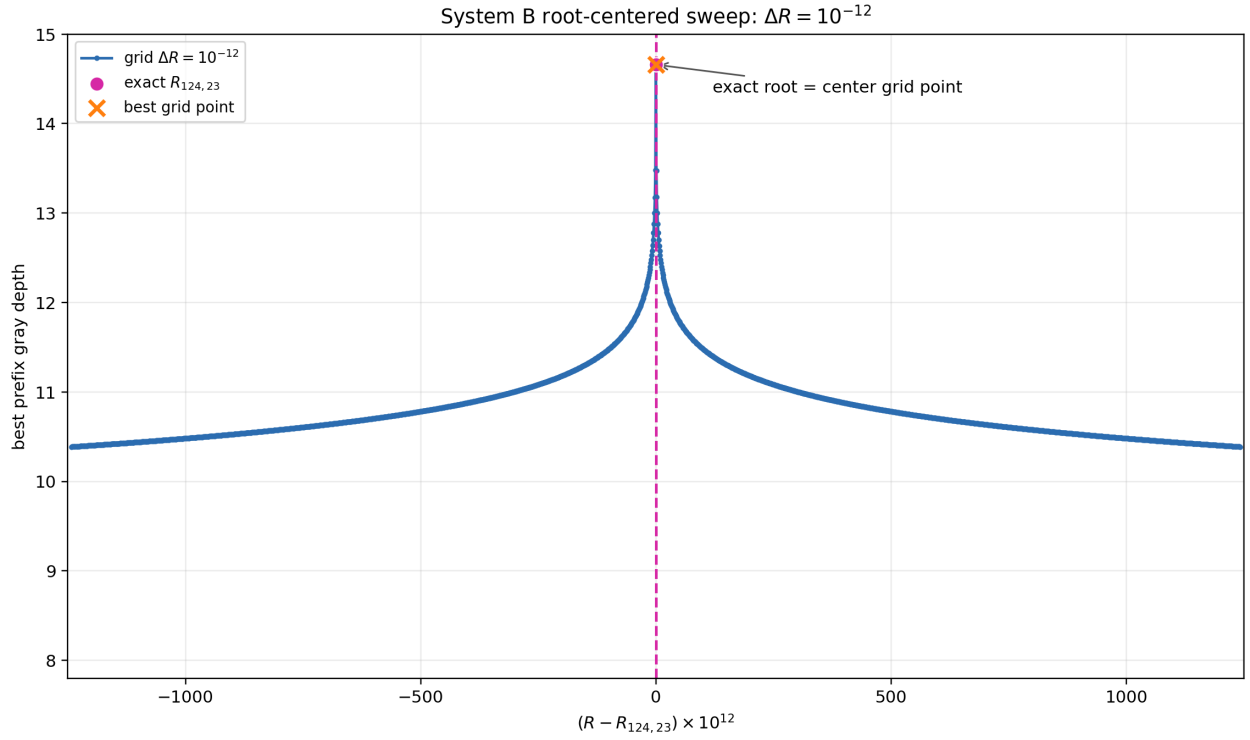


Figure 2. A $\Delta R = 10^{-12}$ sweep with $R_{124,23}$ placed on the central grid point. The deepest point, analytic root, and central grid point coincide. In double precision, the depth is 14.6586 and the error is 2.19×10^{-15} . Reading this root through the correspondence hypothesis $N(R) = 4\pi/(1 - R)^2$ gives

$$N(R_{124,23}) = 137.036042914598545 \dots$$

The CODATA 2022 value is

$$\alpha^{-1}(0) = 137.035999177(21),$$

so the difference is

$$4.37376 \times 10^{-5}.$$

The two values are close but not identical.

4.2 Former High-Energy Candidate: $R_{122,23}$

Candidate 2 was magnified in the same format. Its center is

$$R_{122,23} = \cos^2\left(\frac{23\pi}{122}\right) = 0.688363946817592550 \dots$$

System B: $R_{122,23}$ root-centered resolution comparison

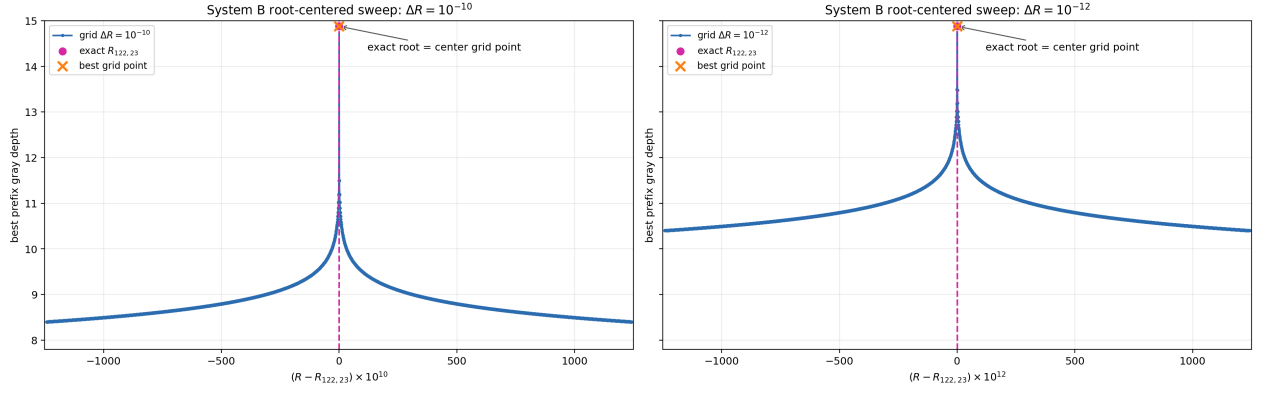


Figure 3. Comparison of $\Delta R = 10^{-10}$ and 10^{-12} sweeps centered on $R_{122,23}$. At both resolutions, the analytic root coincides with the deepest point. The depth is 14.8777 and the error is 1.33×10^{-15} .

The corresponding readout is

$$N(R_{122,23}) = 129.394099680960693 \dots$$

Thus, the direct identity of the second principal peak previously called a candidate near high-energy 128 is the finite-order resonance $R_{122,23}$.

4.3 Near the High-Energy Observed Value: $R_{620,117}$

Higher-order finite-order roots were examined near the R value corresponding to

$$\alpha^{-1}(M_Z^2) = 128.946 \pm 0.015.$$

This search identified the even-order root

$$R_{620,117} = \cos^2\left(\frac{117\pi}{620}\right) = 0.687825191114145187 \dots$$

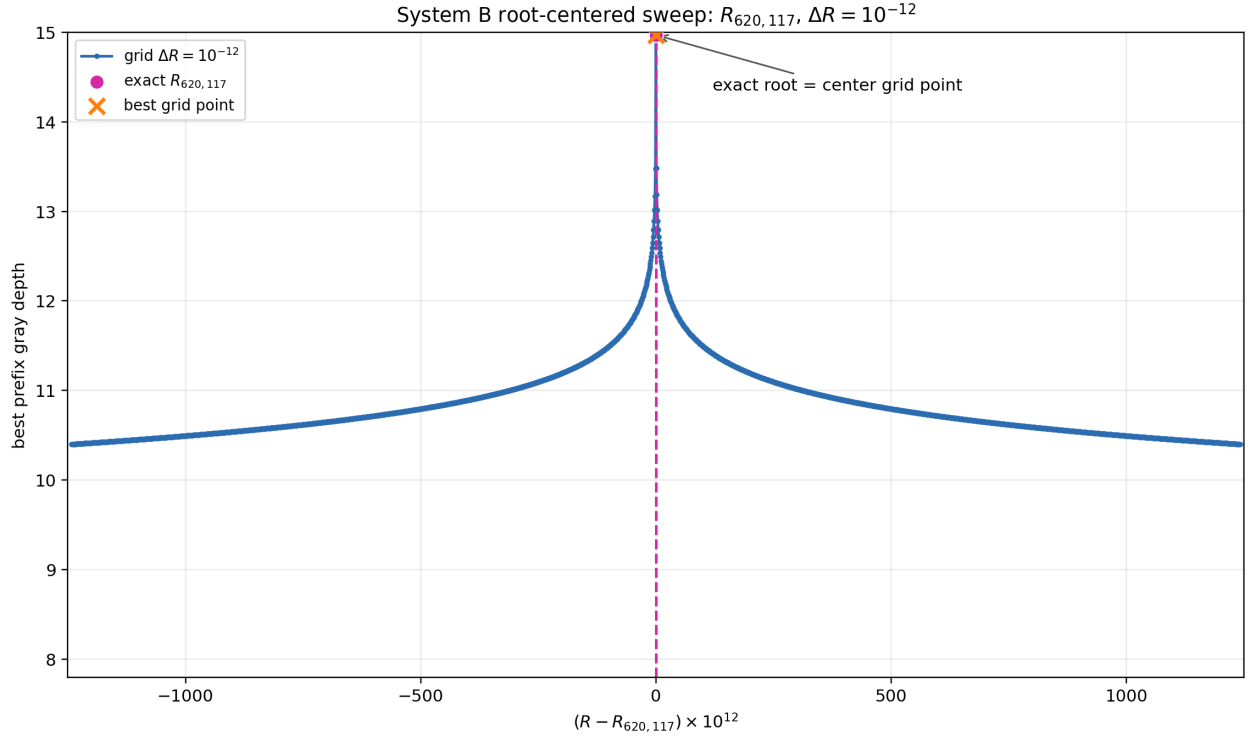


Figure 4. A $\Delta R = 10^{-12}$ sweep with $R_{620,117}$ on the central grid point. The deepest point is the analytic root itself. The depth is 14.9693, the error is 1.07×10^{-15} , and the best prefix ends at $2n - 1 = 1239$.

This root gives

$$N(R_{620,117}) = 128.947864735670559 \dots$$

The difference from the observed value is

$$128.947864735670559 - 128.946 = 0.001864735670559,$$

and hence

$$\frac{0.001864735670559}{0.015} = 0.1243157,$$

or 0.1243σ .

The previous full-range sweep used at most 1024 steps. The gray metric for $R_{620,117}$ requires

$$2n = 1240$$

samples to read two complete periods. Therefore, this root could not appear as a complete peak within the earlier time window. The root near the high-energy observed value was found later not only because of spatial resolution along R , but also because the iteration length in the time direction had been insufficient.

4.4 Odd-Order Control: $R_{567,107}$

In the same observed region,

$$R_{567,107} = 0.687816908934256712 \dots$$

gives

$$N(R_{567,107}) = 128.941022883719000 \dots$$

However, $n = 567$ is odd, so the discrete orbit does not contain the opposite gray-metric extremum -0.02 as a grid point. Thus, even though the operator satisfies

$$U_R^{567} = I,$$

the amplitude readout retains the deficit

$$0.01 \left(1 - \cos \frac{\pi}{567}\right) = 1.5349792919 \times 10^{-7}.$$

The measured deficit is

$$1.5349792966 \times 10^{-7},$$

in agreement with the prediction.

This control shows that finite-order operator recurrence and an infinite-depth peak in the selected observable are not synonymous. Operator recurrence holds regardless of the parity of n , while the present gray-amplitude readout produces a machine-precision peak at even fundamental orders.

5. Multiprecision Calculation and Infinitely Sharp Peaks

5.1 50- and 80-Digit Calculations

We placed $R_{124,23}$ and $R_{620,117}$ on the central grid point and calculated them with 50- and 80-digit arithmetic.

Root	Precision	ΔR	Central depth	Central error	Best step
$R_{124,23}$	50 digits	10^{-14}	52.7036	1.98×10^{-53}	247
$R_{620,117}$	50 digits	10^{-14}	52.6831	2.07×10^{-53}	1239
$R_{124,23}$	80 digits	10^{-16}	82.1106	7.75×10^{-83}	247
$R_{620,117}$	80 digits	10^{-16}	83.0581	8.75×10^{-84}	1239

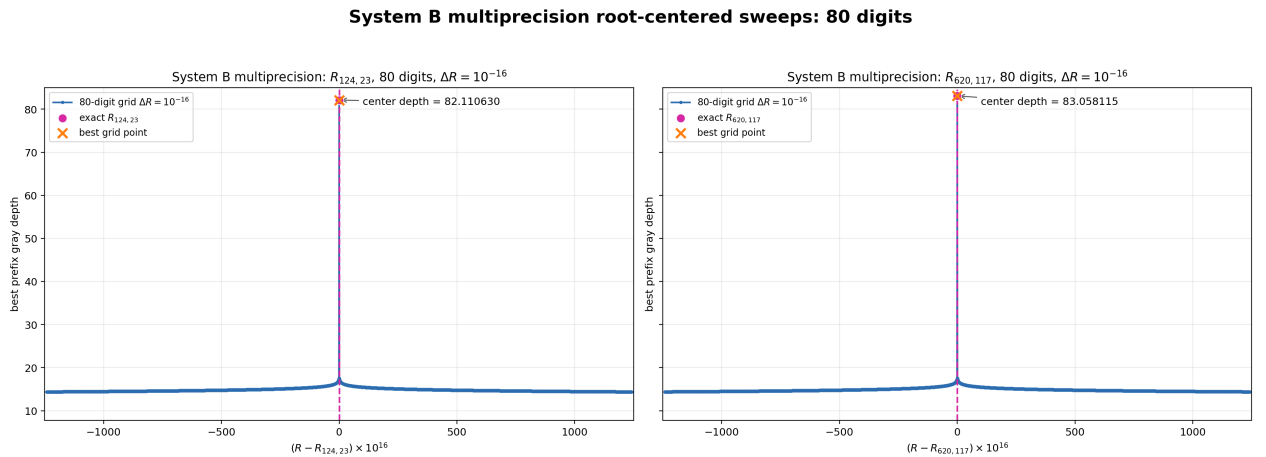


Figure 5. An 80-digit multiprecision sweep of $R_{124,23}$ and $R_{620,117}$. For both roots, the central grid point is the deepest point, and the depth exceeds 80 as the arithmetic precision is increased.

5.2 The Theoretical Limit

At a root whose relatively prime representation (m, n) has even fundamental order n , reading two complete periods $2n$ gives exactly

$$\bar{S} = 0, \quad A = 0.02, \quad D = 0.$$

Therefore,

$$\varepsilon = 0,$$

and

$$d = -\log_{10} \varepsilon \longrightarrow +\infty.$$

The diverging quantity is gray depth, not wave energy or amplitude. The height at the peak center increases with precision while the width narrows. A finite-precision measurement or coarse R sweep can therefore miss the peak easily.

6. Mathematical Origin of the Peaks

6.1 Symmetric and Antisymmetric Decomposition

Decompose the two-channel state as

$$X_j := \frac{A_j + B_j}{\sqrt{2}}, \quad Y_j := \frac{A_j - B_j}{\sqrt{2}}.$$

The eigenvalues of the exchange operator are

$$\lambda_s = r + t = 1,$$

and

$$\lambda_a = r - t = -e^{2i\theta}.$$

Therefore,

$$X_j = X_0, \quad Y_j = \lambda_a^j Y_0.$$

With the projection operators

$$P_s = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_a = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

we obtain

$$\boxed{U_R^j = P_s + \lambda_a^j P_a}.$$

6.2 Derivation of the Finite-Order Roots

Let

$$R_{n,m} = \cos^2 \left(\frac{\pi m}{n} \right), \quad 1 \leq m < \frac{n}{2}, \quad \gcd(m, n) = 1.$$

Then

$$\theta = \arcsin \sqrt{R_{n,m}} = \frac{\pi}{2} - \frac{\pi m}{n},$$

so

$$\lambda_a = -e^{2i\theta} = e^{-2\pi im/n}.$$

It follows that

$$\lambda_a^n = 1$$

and

$$\boxed{U_{R_n, m}^n = I}.$$

This is not a numerical fit. It is an analytic consequence derived directly from the scattering coefficients used in the current program.

6.3 Closed Form of the System B Time Series

The population difference is

$$S_j = \frac{2 \operatorname{Re} (\langle X_0, Y_0 \rangle \lambda_a^j)}{\|X_0\|^2 + \|Y_0\|^2}.$$

Under the principal condition $\phi = 0$ and $s_0 = 0.01$,

$$\frac{2 \langle X_0, Y_0 \rangle}{\|X_0\|^2 + \|Y_0\|^2} = 0.02,$$

and therefore

$$\boxed{S_j = 0.02 \cos \left[j \left(\pi + 2 \arcsin \sqrt{R} \right) \right]}.$$

This equation shows that the iterated scattering in the original program and the Chebyshev recurrence used for the multiprecision calculation evaluate the same time series.

6.4 Low Sensitivity to Initial Values

For a general initial state,

$$S_j = C \cos(j\omega + \varphi_0), \quad \omega = \arg \lambda_a.$$

The amplitude C and initial phase φ_0 depend on the initial state. However,

$$\omega(R) = \pi + 2 \arcsin \sqrt{R}$$

is determined by the operator alone. Consequently, changing the initial amplitude, initial phase, wave-packet shape, or harmonic content does not move

$$R_{n, m}.$$

What changes is the peak height, background, and visibility.

6.5 The Numbers 31, 124, and 248

At $R_{124,23}$,

$$\lambda_a^{31} = i, \quad \lambda_a^{62} = -1, \quad \lambda_a^{93} = -i, \quad \lambda_a^{124} = 1.$$

Thus,

$$31 \times 4 = 124$$

is one complete operator period, and

$$124 \times 2 = 248$$

is the number of samples used by the gray metric to read two complete periods.

The number 248 was not inserted into the scattering coefficients. It appeared as the time window that reads the 124-cycle closed orbit twice.

7. Program Structure

7.1 From Reflection Probability to Complex Scattering Amplitudes

The following excerpt is the core of the minimal System B implementation v5:

```
def scattering_coefficients(reflection_rate: float):
    if reflection_rate < 0.0 or reflection_rate > 1.0:
        raise ValueError("reflection rate must be in [0, 1]")

    delta_f = 2.0 * math.asin(math.sqrt(reflection_rate))
    half_delta = 0.5 * delta_f
    phase = complex(math.cos(half_delta), math.sin(half_delta))

    t = phase * math.cos(half_delta)
    r = -1j * phase * math.sin(half_delta)
    return complex(t), complex(r), abs(t) ** 2, abs(r) ** 2
```

The only input to this function is R . Neither 137, 128, α , $N(R)$, nor a finite-order root enters the calculation of the scattering amplitudes.

Source:

[run_minimal_system_B_gray_direct_check_v5.py](#)

7.2 Scattering Update along the Time Direction

The core operation that generates the peaks is

```
a, b = normalize_pair(
    r * a + t * b,
    t * a + r * b,
)
```

The iterative core is

```
for step in range(steps + 1):
    p_a = abs(a) ** 2
    p_b = abs(b) ** 2
    q = p_a + p_b
    s_value = (p_a - p_b) / q

    # readout of the current state
```

```

prefix_sums.append(prefix_sums[-1] + s_value)

if step >= steps:
    break

a, b = normalize_pair(
    r * a + t * b,
    t * a + r * b,
)

```

Source:

[run_minimal_system_B_gray_direct_check_v5.py](#)

From the second iteration onward, the output A', B' of the preceding scattering event is used as the input to the next application of the same scattering map. In the present code, the variable `step` carries the time-development direction.

7.3 Gray Error

The evaluation function is

```

def prefix_error(prefix_sums, s_min, s_max):
    n = len(prefix_sums) - 1
    total = prefix_sums[n]
    s_mean = total / n
    s_amp = 0.5 * (s_max - s_min)

    mid = n // 2
    first_mean = prefix_sums[mid] / mid
    second_mean = (total - prefix_sums[mid]) / (n - mid)
    s_drift = abs(second_mean - first_mean)

    gray_error = (
        abs(s_mean)
        + abs(s_amp - 0.02)
        + s_drift
    )
    return gray_error, s_mean, s_amp, s_drift

```

Source:

[run_minimal_system_B_gray_direct_check_v5.py](#)

No value 137 or 128 appears here either. The function evaluates only the mean, amplitude, and temporal drift of a periodic orbit.

7.4 Analytic Eigenvalues and Finite-Order Roots

The following analytic code was added after the numerical peaks had been found:

```

def exact_eigenvalues(reflection_rate: float):
    theta = math.asin(math.sqrt(reflection_rate))
    lambda_symmetric = 1.0 + 0.0j
    lambda_antisymmetric = -complex(
        math.cos(2.0 * theta),
        math.sin(2.0 * theta),
    )
    return lambda_symmetric, lambda_antisymmetric

def resonant_R(period_n: int, mode_m: int) -> float:
    return math.cos(math.pi * mode_m / period_n) ** 2

```

Source:

[phase5_eigenphase_resonance_v2.py](#)

This code did not insert the roots in order to create the peaks. It was written afterward to identify the peaks from the full-range sweep as operator-eigenvalue roots.

7.5 Closed-Form Multiprecision Recurrence

Rather than directly evaluating

$$S_j = 0.02 \cos(j\omega),$$

the multiprecision calculation used a Chebyshev recurrence to control rounding error:

```
phase = mp.pi + 2 * mp.asin(mp.sqrt(reflection_rate))
phase_cos = mp.cos(phase)

cosine_previous = mp.mpf(1)
cosine_current = phase_cos

for sample_index in range(sample_count):
    if sample_index == 0:
        cosine_value = cosine_previous
    elif sample_index == 1:
        cosine_value = cosine_current
    else:
        cosine_previous, cosine_current = (
            cosine_current,
            2 * phase_cos * cosine_current - cosine_previous,
        )
        cosine_value = cosine_current

    s_value = mp.mpf("0.02") * cosine_value
```

Source:

[run_two_physical_roots_multiprecision_v1.py](#)

This is not a separate model. It is a closed-form calculation of the same time series derived from the original two-channel exchange update.

7.6 Vector Wave-Packet Update in System A

System A applies the same scattering kernel to wave-packet vectors rather than scalar amplitudes:

```
for collision in range(max_collision + 1):
    rows.append(
        row_for_state(..., "A_channel", a, ...)
    )
    rows.append(
        row_for_state(..., "B_channel", b, ...)
    )

    if collision >= max_collision:
        break

    a_next = src.normalize(r * a + t * b)
    b_next = src.normalize(t * a + r * b)
    a, b = a_next, b_next
```

Source:

[run_system_A_localization_exchange_R_sweep_preliminary_v1.py](#)

Systems A and B use different observables, but their time-development kernel U_R is identical.

7.7 Collision Trigger and the Scope of Iterated Scattering

The complete interaction separates into two stages:

1. The A and B wave packets approach within one cell, triggering the first scattering event.
2. The post-scattering outputs A' , B' are updated into the next cycle of iterated scattering.

The current System A and System B programs directly implement the second stage. They do not implement the trajectories by which wave packets approach through space until the first-collision condition is satisfied.

The result of this paper is therefore not a completed model of the entire interaction process. It is the identification that **the iterated exchange-scattering kernel after collision onset is sufficient to generate finite-order resonance**.

8. Audit for Hidden Parameters

8.1 Classification of the Code Series

The existing code contains three distinct classes with different purposes.

Code class	Known 137/128 values	Role
Early exploration and comparison code	Included	Compare robustness to harmonics and initial conditions at candidate points
Uniform System B v5 sweep	Not included in sweep evaluation	Discover candidate peaks from an equally spaced grid
Analytic-root and multiprecision code	Roots explicit	Verify agreement between discovered peaks and analytic roots

The early System A exploration code and early System B code explicitly contained the 137- and 128-corresponding points as fixed probes. Those calculations alone are therefore not used as evidence of independent discovery.

In contrast, the System B v5 sweep mode generated a Decimal equally spaced sequence from command-line minimum, maximum, and spacing parameters, and applied the same evaluation function at every point:

```
r_values = decimal_range(  
    args.min_R,  
    args.max_R,  
    args.delta_R,  
)  
  
for r_text in r_values:  
    payload = probe_r(  
        float(r_text),  
        r_text,  
        args.steps,  
        args.phi_mode,  
        args.min_steps,  
        args.early_stop_patience,  
    )
```

Source:

[run_minimal_system_B_gray_direct_check_v5.py](#)

A default R for single-point execution remains in the source, but it is not referenced when sweep mode is invoked with `min-R`, `max-R`, and `delta-R`.

8.2 Chronology of Discovery and Verification

The evidential order is:

1. Observe localization transfer and robustness to harmonics in System A.
2. Construct the metastable interface in System B.
3. Perform the full R sweep with the explicit aim of testing the α correspondence.
4. Discover the principal peaks on a 10^{-7} grid.
5. Perform root-centered precision sweeps.
6. Find that the peak centers coincide with $R_{124,23}$ and $R_{122,23}$.
7. Derive the general roots $R_{n,m}$ from the operator eigenvalue.
8. Identify $R_{620,117}$ near the high-energy observed value.
9. Confirm precision-dependent growth of the depth with 50- and 80-digit calculations.
10. Only then search the literature and recognize connections to exact Floquet recurrence, Farey structure, and E_8 .

This chronology establishes that the peaks were not generated by embedding a previously known cyclotomic root or the 31×8 structure of E_8 in the numerical code.

8.3 Reproduction Conditions

The conditions of the full-range sweep are preserved in the CSV file

[high_to_ext_full_delta1e-7_stats_v5.csv](#).

The principal parameters are:

Item	Value
$R_{\min}$	0.68660290255614798
$R_{\max}$	0.70246536756843059
ΔR	10^{-7}
Number of points	158,625
steps	1024
min steps	256
patience	20
phi	0

The inputs, precision, number of points, raw JSON data, and figures for the local and multiprecision sweeps are likewise preserved in their result directories.

9. The Two Physical Models and Resonance Visibility

9.1 Harmonics Do Not Create the Roots; They Make the Roots Visible

Let the internal degrees of freedom of a wave packet be $\mathcal{H}$. Where the same exchange law acts on every internal component,

$$U_R^{\text{total}} = U_R \otimes I_{\mathcal{H}}.$$

The harmonic type, amplitude, phase, and wavelength change the initial vector in $\mathcal{H}$, but they do not change the two-channel eigenvalues

$$1, \quad -e^{2i\theta}.$$

Additional harmonics are therefore not the cause that creates the finite-order roots. They are degrees of freedom that couple the antisymmetric eigensector to localization or population-difference observables and thereby make the resonance visible.

The exact conditions under which the resonance does not appear in the System B population difference are

$$Y_0 = 0$$

or

$$\langle X_0, Y_0 \rangle = 0.$$

In these cases, the antisymmetric eigenphase exists in the operator but is absent from the selected readout.

9.2 Why Different Models Read the Same Region

System A reads localization, harmonic order, and transfer ratio through

$$\mathcal{O}_A(U_R^j \Psi_0).$$

System B reads the A/B population difference, gray error, and metastable depth through

$$\mathcal{O}_B(U_R^j \Psi_0).$$

The observation functions differ, but the time-development operator is the same. The different physical models respond in the same R region because their common exchange-scattering kernel has the same eigenphase.

This is the point of contact with physics in the present study. We did not begin by enumerating abstract cyclotomic roots. Experimental code designed to investigate wave-packet localization transfer and metastable selection detected the same operator resonance.

10. Connections to Prior Work Recognized after the Discovery

10.1 Chronology of the Literature Search

The works discussed below did not guide the full-range sweep or the program design. We searched the literature only after identifying $R_{124,23}$, $R_{122,23}$, and $R_{620,117}$ through numerical calculation and eigenvalue analysis.

This chronology is stated explicitly for two reasons:

1. to show that the numerical experiment was not a reproduction obtained by embedding known cyclotomic roots or an E_8 decomposition in the code; and
2. to distinguish what is known as a general phenomenon from the mathematical model and numerical discovery specific to this study.

10.2 Exact State-Independent Quantum Recurrence

Anand et al. classified exact state-independent recurrence times from the cyclotomic spectrum of finite-dimensional Floquet unitary operators [5]. The present relation

$$U_R^n = I$$

is a concrete two-channel instance connected to this general class of exact recurrence.

We therefore do not claim to have discovered the general concept of a finite-order unitary operator. The contribution is that a wave-packet numerical experiment aimed at the fine-structure constant independently reached this structure and detected it in both localization-transfer and metastable-interface models.

10.3 Farey Sequences and Resonance Organization

Tomás described the organization of physical resonance diagrams by Farey sequences of rational numbers [6]. The present roots are classified by

$$\frac{m}{n}.$$

Farey adjacency can therefore organize the closely spaced high-order roots on the high-energy side. That work, however, does not treat the exchange-scattering map, the fine-structure-constant correspondence, localization transfer, or the cat model used here.

10.4 The 31×8 Structure of E_8

After obtaining

$$124 = 31 \times 4, \quad 248 = 31 \times 4 \times 2,$$

we investigated a possible connection to the 248 dimensions of E_8 and its decomposition into 31 orthogonal eight-dimensional subspaces.

Thompson and later work treat structures associated with a decomposition of E_8 into 31 orthogonal eight-dimensional Cartan subalgebras [7,8]. The E_8 root system also separates into 112 integer-type roots and 128 half-integer-type roots [9].

These connections were noticed only after the finite-order-root discovery of July 18, 2026. The numbers 31, 128, and 248 were not selected in advance to tune the iteration count or exchange coefficient.

At present, no explicit map has been constructed from

$$31 \text{ time sectors} \times 4 \text{ phases} \times 2 \text{ channels}$$

to the 31 eight-dimensional subspaces of E_8 . The E_8 connection is therefore retained as a structural hypothesis.

10.5 Exceptional Algebras and the Fine-Structure Constant

El Naschie proposed a derivation connecting the $8 \times 31 = 248$ structure of E_8 to the fine-structure constant [10]. Singh discussed $1/137$ using octonionic spacetime and an exceptional Jordan algebra [11].

Neither line of work was the starting point of this study. We deliberately searched for the fine-structure constant, but did not use E_8 or an exceptional algebra to design the numerical peaks. Their relation to the present result was investigated only after obtaining the finite-order roots and the $31 \times 4 \times 2$ structure.

11. Discussion

11.1 What Did We Discover?

We initially expected that a closed exchange system could directly explain the fine-structure constant.

The numerical experiments did return sharp peaks near 137 and 128. Their direct cause, however, was not the assumed electromagnetic-coupling correspondence

$$G_R = e$$

itself. The direct cause is the finite-order exchange resonance

$$\lambda_a^n = 1.$$

Thus, this study has not completed a direct derivation of α . In the course of searching for the fine-structure constant, it discovered that the iterated exchange-scattering program detects exact finite-order recurrence.

Because general exact recurrence is present in prior work, the general phenomenon is a rediscovery of a different phenomenon. The following points are nevertheless specific to this study:

- it began from a wave-packet localization-transfer model;
- it also began from a weak-readout/strong-selection model of two metastable states;
- both models share the same exchange-scattering kernel;
- multiple even-order roots appear near the physical readout regions 137 and 128; and
- source code, raw data, full-range figures, local figures, and multiprecision results are preserved in reproducible form.

11.2 Why Are the Roots Near α ?

This is the most important question remaining after the result of this paper.

On the low-energy side,

$$N(R_{124,23}) = 137.036042914598545 \dots$$

is close to

$$\alpha^{-1}(0) = 137.035999177(21).$$

On the high-energy side,

$$N(R_{620,117}) = 128.947864735670559 \dots$$

is close to

$$\alpha^{-1}(M_Z^2) = 128.946 \pm 0.015.$$

There are countably infinitely many finite-order roots

$$R_{n,m} = \cos^2\left(\frac{\pi m}{n}\right).$$

Their arithmetic density can place high-order roots near an arbitrary real value and may account for part of the proximity.

However, this study did not merely select arbitrary roots after the fact:

- the original full-range sweep produced $R_{124,23}$ and $R_{122,23}$ as its principal peaks;
- the low-energy root has the short 31×4 temporal structure;
- an even root $R_{620,117}$ with infinite depth under the present gray metric lies near the high-energy observed value; and
- different wave-packet models responded to the same exchange-scattering region.

The proximity therefore cannot yet be declared either a mere coincidence or a physical necessity.

11.3 The Central Open Problem

The central open problem of this paper is stated without ambiguity:

We began this search expecting that a closed exchange system could explain the fine-structure constant α . Within what has been established by the present analysis, however, we did not derive α itself; we merely rediscovered a different finite-order resonance phenomenon connected to the general theory of exact recurrence. Yet the extremely sharp peaks of that finite-order resonance actually occur near the two physical low- and high-energy values of α^{-1} . Why this proximity occurs remains unexplained. Is it an accidental consequence of arithmetic density, or is it another necessity imposed by four-dimensional discrete geometry, E_8 -type symmetry, a physical selection rule of exchange scattering, or an as-yet-unidentified structure? Distinguishing these alternatives is the central task of future work.

The required discriminating calculations are:

1. enumerate all roots below a fixed maximum order $n_{\max}$;
2. calculate how often an even-order root at least as close to the observed value occurs near randomized reference values;
3. test whether localization, entropy, phase correlation, and transfer observables select the same roots independently of the gray metric;
4. randomize the initial state, harmonics, and time window to separate root location from visibility;
5. construct an explicit map connecting the four-dimensional 137-cell count, the 128 outer sector, the 31×8 decomposition, and the exchange operator; and
6. implement the same U_R in a physical optical coupler, discrete-time quantum walk, or two-level Floquet system.

12. Conclusion

This study identified the direct cause of the sharp peaks observed near the fine-structure-constant inverse values 137 and 128.

The peaks form where the antisymmetric eigenvalue

$$\lambda_a = -e^{2i \arcsin \sqrt{R}}$$

of

$$U_R = \begin{pmatrix} r & t \\ t & r \end{pmatrix}$$

has finite order, namely at

$$R_{n,m} = \cos^2 \left(\frac{\pi m}{n} \right).$$

The principal low-energy peak is $R_{124,23}$, the former high-energy principal candidate is $R_{122,23}$, and $R_{620,117}$ lies near the high-energy observed value.

With 50- and 80-digit arithmetic, the depth increases with numerical precision. In ideal arithmetic, the gray error vanishes at an even fundamental-order root and the resonance depth becomes infinite.

The result is not the first presentation of general quantum recurrence. It is the discovery that a wave-packet numerical experiment deliberately aimed at the fine-structure constant detects finite-order roots without inserting the known values into its scattering kernel, and that the phenomenon can be verified in reproducible localization-transfer and metastable two-state models.

The fine-structure constant itself remains unexplained. A new question has been isolated: why do extremely sharp finite-order roots occur near the two physical values of α^{-1} ?

The next stage is to distinguish

coincidence from another necessity.

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Self-References

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Appendix A. Principal Reproduction Files

Content	File
July 15, 2026 complete paper	Previous paper, English
Full-range sweep statistics	Full-range sweep CSV
Full-range sweep figure	Figure 1 PNG
$R_{124,23}$ local sweep	Local-sweep result
$R_{122,23}$ resolution comparison	Resolution-comparison result
Higher-order-root comparison	Higher-order-root result
Multiprecision result	Multiprecision result
Precision-matched ultralocal sweep	Ultralocal-sweep result
Minimal System B scattering kernel	System B source
Eigenvalue and analytic-root code	Eigenvalue-analysis source
Multiprecision code	Multiprecision source

Appendix B. Classification of Claims

Claim	Classification
Unitarity of U_R	Derived consequence
$\lambda_s = 1, \lambda_a = -e^{2i\theta}$	Derived consequence
$R_{n,m} = \cos^2(\pi m/n)$	Derived consequence
$U_{R_{n,m}}^n = I$	Derived consequence
Infinite ideal gray depth at even fundamental-order roots	Derived consequence of the present readout
Agreement of the full-range peaks with $R_{124,23}$ and $R_{122,23}$	Numerical fact
Proximity of $R_{620,117}$ to the high-energy observed value	Numerical fact
Harmonics control visibility rather than root position	Theoretical insight from the operator decomposition
Identity with the 31×8 decomposition of E_8	Structural hypothesis
Necessity of the proximity between finite-order roots and physical α	Not derived; central open problem